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Azure Lab : #20

Step by step process :

```
nur [ ~ ]$ az vm create \
  --resource-group "learn-eb1d0f58-e002-4b9c-a4d2-df135b17
5af2" \
  --name MeanStack \
  --image Canonical:0001-com-ubuntu-server-focal:20_04-lts
:latest \
  --admin-username azureuser \
  --generate-ssh-keys
SSH key files '/home/nur/.ssh/id_rsa' and '/home/nur/.ssh/
id_rsa.pub' have been generated under ~/.ssh to allow SSH
access to the VM. If using machines without permanent stor
age, back up your keys to a safe location.
Consider upgrading security for your workloads using Azure
Trusted Launch VMs. To know more about Trusted Launch, pl
ease visit https://aka.ms/TrustedLaunch.
```

```
{
  "fqdns": "",
  "id": "/subscriptions/66934742-a2f4-4f94-ad7b-07a47d83d2bc/resourceGroups/learn-eb1d0f58-e002-4b9c-a4d2-df135b175af2/providers/Microsoft.Compute/virtualMachines/MeanStack",
  "location": "westus",
  "macAddress": "60-45-BD-04-E3-65",
  "powerState": "VM running",
  "privateIpAddress": "10.0.0.4",
  "publicIpAddress": "52.160.104.105",
  "resourceGroup": "learn-eb1d0f58-e002-4b9c-a4d2-df135b175af2",
  "zones": ""
}
```

```
nur [ ~ ]$ az vm open-port \
  --port 80 \
  --resource-group "learn-eb1d0f58-e002-4b9c-a4d2-df135b175af2" \
  --name MeanStack
```

Output

```
{
  "defaultSecurityRules": [
    {
      "access": "Allow",
      "description": "Allow inbound traffic from all VMs in VNET",
      "destinationAddressPrefix": "VirtualNetwork",
      "destinationAddressPrefixes": [],
      "destinationPortRange": "*",
      "destinationPortRanges": [],
      "direction": "Inbound",
      "etag": "W/\"2d3c8c4e-3603-41d7-8ca4-a3bdd8b697b0\"",
      "id": "/subscriptions/66934742-a2f4-4f94-ad7b-07a47d83d2bc/resourceGroups/learn-eb1d0f58-e002-4b9c-a4d2-
```

df135b175af2/providers/Microsoft.Network/networkSecurityGroups/MeanStackNSG/defaultSecurityRules/AllowVnetInBound",

"name": "AllowVnetInBound",

"priority": 65000,

"protocol": "*",

"provisioningState": "Succeeded",

"resourceGroup": "learn-eb1d0f58-e002-4b9c-a4d2-df135b175af2",

"sourceAddressPrefix": "VirtualNetwork",

"sourceAddressPrefixes": [],

"sourcePortRange": "*",

"sourcePortRanges": [],

"type": "Microsoft.Network/networkSecurityGroups/defaultSecurityRules"

},

{

"access": "Allow",

"description": "Allow inbound traffic from azure load balancer",

"destinationAddressPrefix": "*",

"destinationAddressPrefixes": [],

"destinationPortRange": "*",

"destinationPortRanges": [],

"direction": "Inbound",

"etag": "W/\"2d3c8c4e-3603-41d7-8ca4-a3bdd8b697b0\"",

"id": "/subscriptions/66934742-a2f4-4f94-ad7b-07a47d83d2bc/resourceGroups/learn-eb1d0f58-e002-4b9c-a4d2-

df135b175af2/providers/Microsoft.Network/networkSecurityGroups/MeanStackNSG/defaultSecurityRules/AllowAzureLoadBalancerInBound",

"name": "AllowAzureLoadBalancerInBound",

"priority": 65001,

"protocol": "*",

```
"provisioningState": "Succeeded",
"resourceGroup": "learn-eb1d0f58-e002-4b9c-a4d2-df135b175af2",
"sourceAddressPrefix": "AzureLoadBalancer",
"sourceAddressPrefixes": [],
"sourcePortRange": "*",
"sourcePortRanges": [],
"type": "Microsoft.Network/networkSecurityGroups/defaultSecurityRules"
},
{
  "access": "Deny",
  "description": "Deny all inbound traffic",
  "destinationAddressPrefix": "*",
  "destinationAddressPrefixes": [],
  "destinationPortRange": "*",
  "destinationPortRanges": [],
  "direction": "Inbound",
  "etag": "W/\"2d3c8c4e-3603-41d7-8ca4-a3bdd8b697b0\"",
  "id": "/subscriptions/66934742-a2f4-4f94-ad7b-07a47d83d2bc/resourceGroups/learn-eb1d0f58-e002-4b9c-a4d2-df135b175af2/providers/Microsoft.Network/networkSecurityGroups/MeanStackNSG/defaultSecurityRules/DenyAllInBound",
  "name": "DenyAllInBound",
  "priority": 65500,
  "protocol": "*",
  "provisioningState": "Succeeded",
  "resourceGroup": "learn-eb1d0f58-e002-4b9c-a4d2-df135b175af2",
  "sourceAddressPrefix": "*",
  "sourceAddressPrefixes": [],
  "sourcePortRange": "*",
```

```
"sourcePortRanges": [],  
"type": "Microsoft.Network/networkSecurityGroups/defaultSecurityRules"  
},  
{  
"access": "Allow",  
"description": "Allow outbound traffic from all VMs to all VMs in VNET",  
"destinationAddressPrefix": "VirtualNetwork",  
"destinationAddressPrefixes": [],  
"destinationPortRange": "*",  
"destinationPortRanges": [],  
"direction": "Outbound",  
"etag": "W/\"2d3c8c4e-3603-41d7-8ca4-a3bdd8b697b0\"",  
"id": "/subscriptions/66934742-a2f4-4f94-ad7b-07a47d83d2bc/resourceGroups/learn-  
eb1d0f58-e002-4b9c-a4d2-  
df135b175af2/providers/Microsoft.Network/networkSecurityGroups/MeanStackNSG/d  
efaultSecurityRules/AllowVnetOutBound",  
"name": "AllowVnetOutBound",  
"priority": 65000,  
"protocol": "*",  
"provisioningState": "Succeeded",  
"resourceGroup": "learn-eb1d0f58-e002-4b9c-a4d2-df135b175af2",  
"sourceAddressPrefix": "VirtualNetwork",  
"sourceAddressPrefixes": [],  
"sourcePortRange": "*",  
"sourcePortRanges": [],  
"type": "Microsoft.Network/networkSecurityGroups/defaultSecurityRules"  
},  
{  
"access": "Allow",
```

```
"description": "Allow outbound traffic from all VMs to Internet",
"destinationAddressPrefix": "Internet",
"destinationAddressPrefixes": [],
"destinationPortRange": "*",
"destinationPortRanges": [],
"direction": "Outbound",
"etag": "W/\"2d3c8c4e-3603-41d7-8ca4-a3bdd8b697b0\"",
"id": "/subscriptions/66934742-a2f4-4f94-ad7b-07a47d83d2bc/resourceGroups/learn-
eb1d0f58-e002-4b9c-a4d2-df135b175af2/providers/Microsoft.Network/networkSecurityGroups/MeanStackNSG/defaultSecurityRules/AllowInternetOutBound",
"name": "AllowInternetOutBound",
"priority": 65001,
"protocol": "*",
"provisioningState": "Succeeded",
"resourceGroup": "learn-eb1d0f58-e002-4b9c-a4d2-df135b175af2",
"sourceAddressPrefix": "*",
"sourceAddressPrefixes": [],
"sourcePortRange": "*",
"sourcePortRanges": [],
"type": "Microsoft.Network/networkSecurityGroups/defaultSecurityRules"
},
{
"access": "Deny",
"description": "Deny all outbound traffic",
"destinationAddressPrefix": "*",
"destinationAddressPrefixes": [],
"destinationPortRange": "*",
"destinationPortRanges": [],
```

```
"direction": "Outbound",

"etag": "W/\"2d3c8c4e-3603-41d7-8ca4-a3bdd8b697b0\"",

"id": "/subscriptions/66934742-a2f4-4f94-ad7b-07a47d83d2bc/resourceGroups/learn-
eb1d0f58-e002-4b9c-a4d2-
df135b175af2/providers/Microsoft.Network/networkSecurityGroups/MeanStackNSG/defaultSecurityRules/DenyAllOutBound",

"name": "DenyAllOutBound",

"priority": 65500,

"protocol": "*",

"provisioningState": "Succeeded",

"resourceGroup": "learn-eb1d0f58-e002-4b9c-a4d2-df135b175af2",

"sourceAddressPrefix": "*",

"sourceAddressPrefixes": [],

"sourcePortRange": "*",

"sourcePortRanges": [],

"type": "Microsoft.Network/networkSecurityGroups/defaultSecurityRules"
}

],

"etag": "W/\"2d3c8c4e-3603-41d7-8ca4-a3bdd8b697b0\"",

"id": "/subscriptions/66934742-a2f4-4f94-ad7b-07a47d83d2bc/resourceGroups/learn-
eb1d0f58-e002-4b9c-a4d2-
df135b175af2/providers/Microsoft.Network/networkSecurityGroups/MeanStackNSG",

"location": "westus",

"name": "MeanStackNSG",

"networkInterfaces": [

{

"id": "/subscriptions/66934742-a2f4-4f94-ad7b-07a47d83d2bc/resourceGroups/learn-
eb1d0f58-e002-4b9c-a4d2-
df135b175af2/providers/Microsoft.Network/networkInterfaces/MeanStackVMNic",

"resourceGroup": "learn-eb1d0f58-e002-4b9c-a4d2-df135b175af2"
```

```
}  
],  
"provisioningState": "Succeeded",  
"resourceGroup": "learn-eb1d0f58-e002-4b9c-a4d2-df135b175af2",  
"resourceGuid": "c6f26e6b-a8c0-4ff8-a1aa-4265dbb329e4",  
"securityRules": [  
  {  
    "access": "Allow",  
    "destinationAddressPrefix": "*",  
    "destinationAddressPrefixes": [],  
    "destinationPortRange": "22",  
    "destinationPortRanges": [],  
    "direction": "Inbound",  
    "etag": "W/\"2d3c8c4e-3603-41d7-8ca4-a3bdd8b697b0\"",  
    "id": "/subscriptions/66934742-a2f4-4f94-ad7b-07a47d83d2bc/resourceGroups/learn-  
eb1d0f58-e002-4b9c-a4d2-  
df135b175af2/providers/Microsoft.Network/networkSecurityGroups/MeanStackNSG/s  
ecurityRules/default-allow-ssh",  
    "name": "default-allow-ssh",  
    "priority": 1000,  
    "protocol": "Tcp",  
    "provisioningState": "Succeeded",  
    "resourceGroup": "learn-eb1d0f58-e002-4b9c-a4d2-df135b175af2",  
    "sourceAddressPrefix": "*",  
    "sourceAddressPrefixes": [],  
    "sourcePortRange": "*",  
    "sourcePortRanges": [],  
    "type": "Microsoft.Network/networkSecurityGroups/securityRules"  
  },  
]
```



```
{
  "access": "Allow",
  "destinationAddressPrefix": "*",
  "destinationAddressPrefixes": [],
  "destinationPortRange": "80",
  "destinationPortRanges": [],
  "direction": "Inbound",
  "etag": "W/\"2d3c8c4e-3603-41d7-8ca4-a3bdd8b697b0\"",
  "id": "/subscriptions/66934742-a2f4-4f94-ad7b-07a47d83d2bc/resourceGroups/learn-eb1d0f58-e002-4b9c-a4d2-df135b175af2/providers/Microsoft.Network/networkSecurityGroups/MeanStackNSG/securityRules/open-port-80",
  "name": "open-port-80",
  "priority": 900,
  "protocol": "*",
  "provisioningState": "Succeeded",
  "resourceGroup": "learn-eb1d0f58-e002-4b9c-a4d2-df135b175af2",
  "sourceAddressPrefix": "*",
  "sourceAddressPrefixes": [],
  "sourcePortRange": "*",
  "sourcePortRanges": [],
  "type": "Microsoft.Network/networkSecurityGroups/securityRules"
},
{
  "tags": {},
  "type": "Microsoft.Network/networkSecurityGroups"
}
```

```
nur [ ~ ]$ ipaddress=$(az vm show \
  --name MeanStack \
  --resource-group "learn-eb1d0f58-e002-4b9c-a4d2-df135b175af2" \
  --show-details \
  --query [publicIps] \
  --output tsv)
nur [ ~ ]$ ssh azureuser@$ipaddress
```

Output :

The authenticity of host '52.160.104.105 (52.160.104.105)' can't be established.

ED25519 key fingerprint is SHA256:XiWny4ucgVpK3wnMz6MwHIPfdIsVRfTBclzjlq56ZEY.

This key is not known by any other names.

Are you sure you want to continue connecting (yes/no/[fingerprint])? yes

Warning: Permanently added '52.160.104.105' (ED25519) to the list of known hosts.

Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.15.0-1087-azure x86_64)

* Documentation: <https://help.ubuntu.com>

* Management: <https://landscape.canonical.com>

* Support: <https://ubuntu.com/pro>

System information as of Sat May 3 07:30:39 UTC 2025

System load: 0.0 Processes: 119

Usage of /: 7.2% of 28.89GB Users logged in: 0

Memory usage: 11% IPv4 address for eth0: 10.0.0.4

Swap usage: 0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.

See <https://ubuntu.com/esm> or run: `sudo pro status`

The list of available updates is more than a week old.

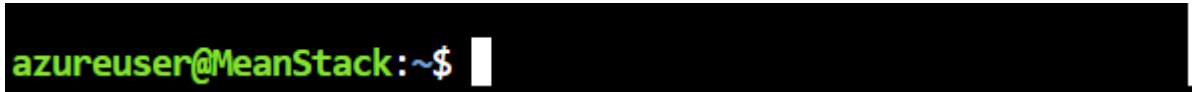
To check for new updates run: `sudo apt update`

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in `/usr/share/doc/*/copyright`.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "`sudo <command>`".

See "`man sudo_root`" for details.

A terminal window with a black background. The prompt is 'azureuser@MeanStack:~\$' in green text. A white cursor is positioned at the end of the prompt.

```
azureuser@MeanStack:~$
```